

Build a screen silhouette

Learning intention

Explain that a shadow forms where an object blocks light from a source.

You need a clear dark shape.

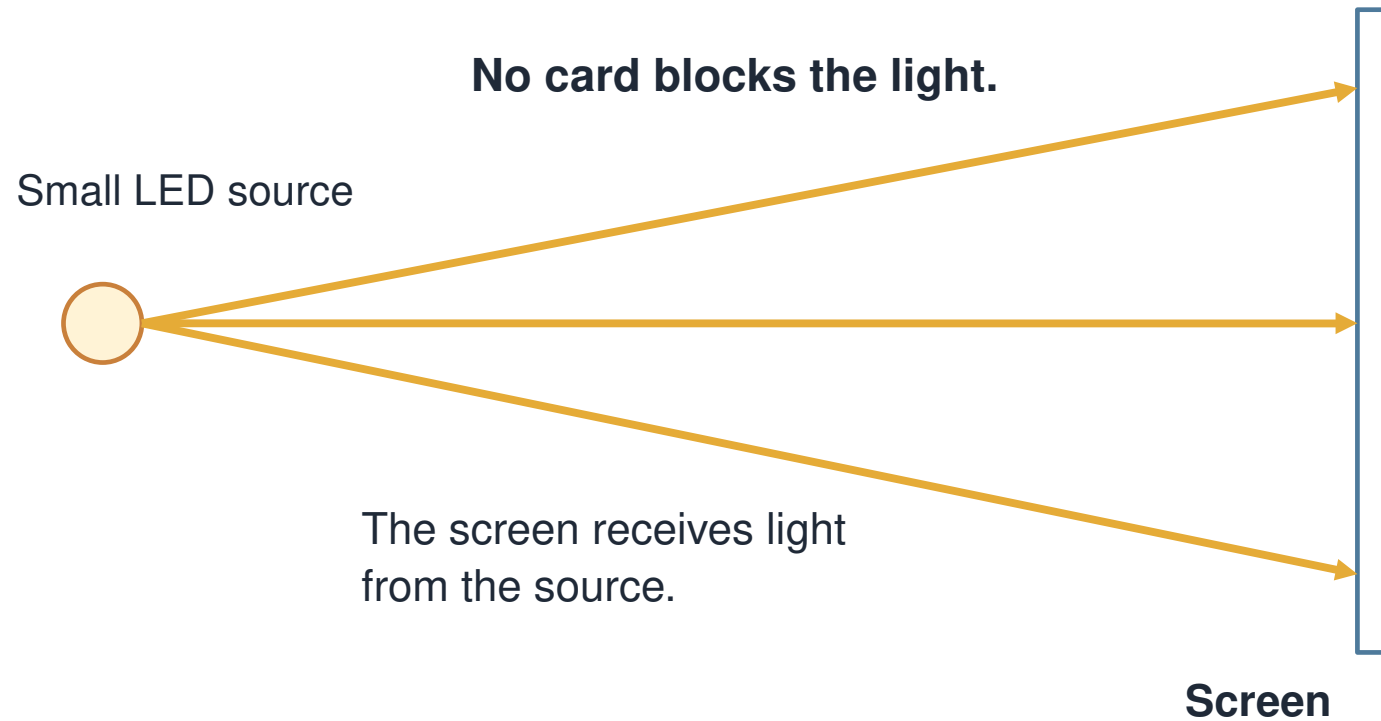
Which material should you choose?

Use the light evidence.

Three clues for our design

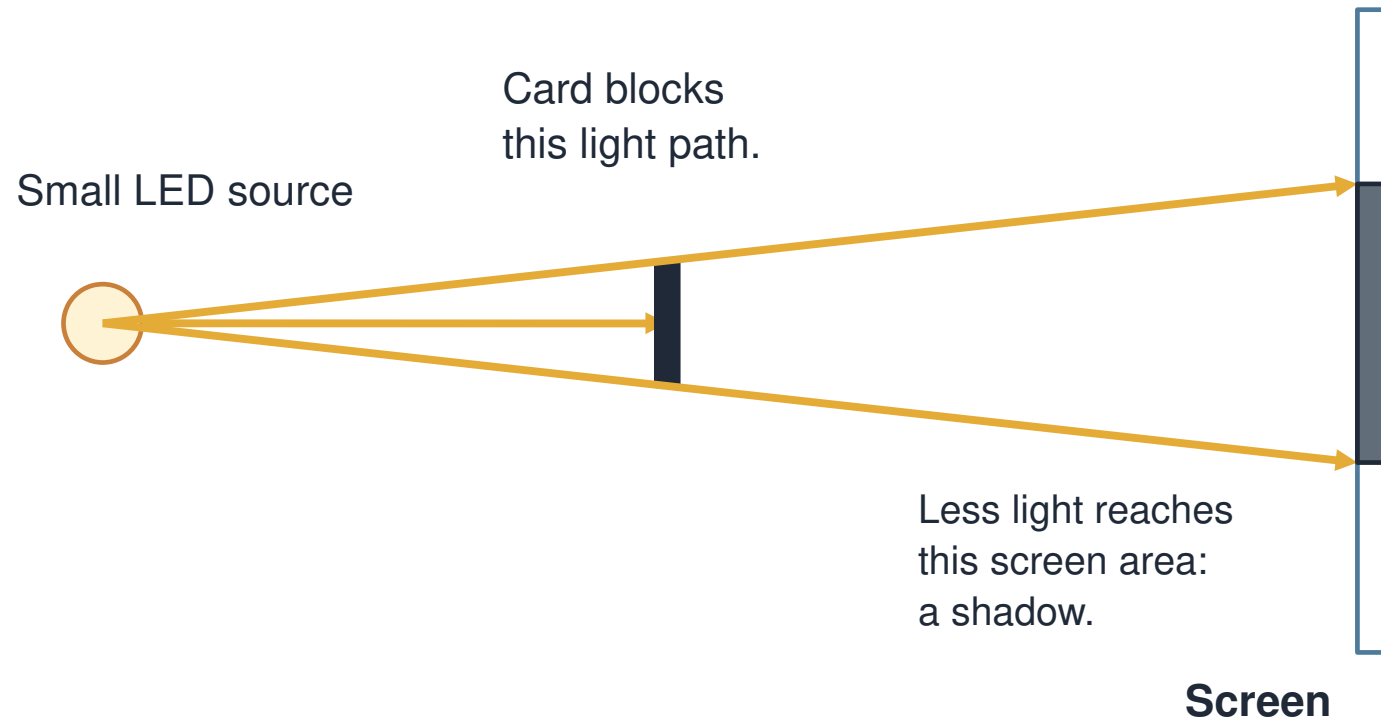
1. Name a light source.
2. What does a surface do when it reflects?
3. Can a card block light?

I start without a blocker



The source sends light to the screen. Light reaches the tested area.

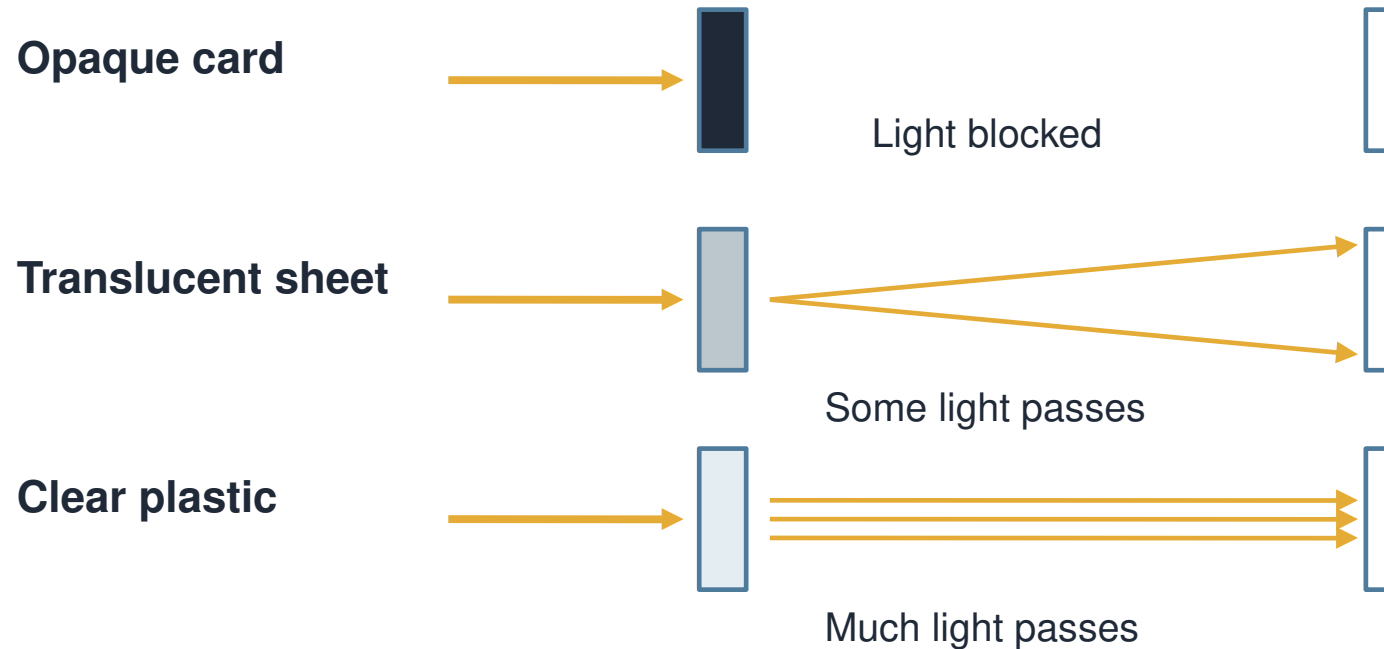
I put a card in the path



The opaque card blocks light. Less reaches part of the screen. That darker area is the shadow.

Materials let different amounts through

Same source, screen and sample position



Opaque: light is blocked. Translucent: some light passes. Clear: much light passes.

Choose a silhouette material

Read the test cards. Choose for a dark silhouette. Point to the evidence.

Card	Sample	Constructed observation
A	Opaque card	Clear dark shape
B	Translucent paper	Lighter, less distinct shape
C	Clear plastic	Most light passes; faint edge visible

Constructed teaching evidence

W1. Which sample fits our design goal?

The disappearing-shape challenge

The card moves out of the beam.

The source stays on.

Predict what changes on the screen.

W2. What explains the change?

Why is there a shadow?

Choose the account that follows the light.

- A** The card sends dark paint onto the screen.
- B** The card blocks light, so less reaches part of the screen.
- C** The source switches off when the card arrives.

Choose your investigation route

Choose and order the setup.

Record a design decision.

Repair a shadow explanation.

Use the evidence

Read or point to one clue.

Record what it supports.

Keep your explanation short.

Make your own science record

Complete your chosen route.

Use a diagram, words or speech.

Check one answer with the evidence.

Check and improve

Does my answer match the evidence?

Have I explained why?

Improve one label or sentence.

Lundy loop: improve our account

Choose how to share one idea.

Listen to another explanation.

Use evidence to improve our account.

What is a shadow made of?

Three quick checks

E1. What does an opaque card do to light?

E2. Where is the shadow in our setup?

E3. What changes when the card leaves the beam?